GLIMPSE: GLobal Isotope Mapping and Precipitation Simulation Experiment

Part I

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Highlights:

- Spatiotemporal connectivity of water isotopes between sink vapour and surface waters
- Identification of moisture pathways using atmospheric river channels deduced from TPWV flow-accumulation gradients.
- Validation of source moisture via seasonal clustering and enroute GNIP records.

Abstract

We investigate the precipitation histories in montane geographies by the isotopic disentanglement of moisture sources with a newly devised, physics-based, machine learning-powered architecture, utilising satellite-derived and in-situ hydrological drivers. GLIMPSE (GLobal Isotope Mapping and Precipitation Simulation Experiment) simulates the hydrological cycle at any spacetime coordinate by optimising the isotopic signature of meteoric waters sourced from snow, rain, glacier melt, groundwater, and evapotranspiration, employing a 1-D moisture back-propagation schema that tracks the isotopic composition of surface waters back to its origin vapour. The schema works on a moisture back trajectory method where Atmospheric Rivers are traced from daily Total Precipitable Water Vapour gradients directed towards the sink of local precipitation. The source vapour composition is then calculated and resolved into open water evaporation, canopy water evaporation, direct evaporation from bare soil, snow evaporation and transpiration inventories. Early runs of the model simulating precipitation over the eastern Himalayas (a) describe spatiotemporal connectivity of water isotopes between sink vapour and surface waters, (b) identify 6 moisture pathways arising from isotopically distinct sources, (c) quantify the effects of spatial and temporal signatures on source vapour isotope composition and (d) link source moisture i.e. evapotranspiration inventory with modelled source vapour isotope composition. This study elucidates the relative importance of seasonality and geography in tracing source vapour to precipitation events in lieu of multi-modal source scenarios.

Keywords: Stable Isotope, Remote Sensing, Machine Learning, Circulation Model, Climate Simulation, Himalayas

1 Introduction

Climate change is modifying global precipitation patterns and bringing about unprecedented changes in the different facets of the water cycle (Trenberth, 2011). In order to be better prepared for the potentially adverse impacts of climate change on water resources, we need to improve our understanding of the water cycle. Stable isotopes of oxygen and hydrogen in water provide a useful medium to disentangle the complex web of Earth System processes. Their natural abundance in meteoric waters makes them an ideal environmental tracer to track the journey of a water particle along its entire hydrologic life cycle. Conversely, the synoptic variability of the water cycle can be used to examine longer-term mean isotope signals.

Figure 1 shows a simplified rendition of the water cycle. Water molecules in the form of vapour are generated over oceans, seas, and terrestrial water bodies and are directed across atmospheric pressure belts under the influence of wind, depending on the season and geography. Moving landward onto cooler atmospheric temperatures, some of this vapour condenses in the form of a rainout (Rozanski et al. 1993). Water molecules containing ^{2}H or ^{18}O preferentially condensate over ^{1}H or ^{18}O-containing vapour, leaving the remaining body of moisture isotopically lighter or depleted in heavy isotopes (Criss, 1999). The longer (*continental effect*: Dansgaard, 1964) or higher (*orographic effect*: Poage and Chamberlin, 2001) the vapour travels, the greater the degree of rainout and, thus, the more depleted the isotopes in precipitation downstream will be, with lower $\delta^{18}O$ and $\delta^{2}H$ values. In principle, variations in $\delta^{18}O$ and $\delta^{2}H$ values are coupled: under conditions of isotopic equilibrium, the rate of fractionation of the heavier molecule w.r.t the lighter molecule for both species is identical and proportional by a factor of eight, and if the conditions are non-equilibrium due to the kinetics associated with temperature or vapour pressure, the proportionality in the fractionation differs from eight (Gat, 1996). Based on the Rayleigh model, the progressive rainout process can be isotopically mapped as a function of transport and phase change

during snow/rain formation in clouds, raindrop re-evaporation and continental recycling, as well as in surface water inventories that are in a state of dynamic mixing as they feed into the meteoric reservoir, which then drains back into the oceans (Rozanski et al., 1993; Ciais and Jouzel 1994; Ciais et al. 1995; Gat, 1996).

Given the usefulness of water isotopes in hydrological and atmospheric applications, a comprehensive understanding of the water cycle is pivotal. Towards this, attempts have been made to model the water cycle as part of the general atmospheric circulation for decades, improving accuracy and magnitude with a better understanding (Joussaume et al. 1984). Aiming to enhance the understanding of stable water isotope processes on local to global scales, current stable isotope models attempt to synthesise the available knowledge on stable isotope processes to various degrees. From the earliest Rayleigh distillation models, research has advanced to Atmospheric General Circulation Models (AGCM), which have stable isotope processes coupled into the model's hydrological cycle (Joussaume et al. 1984). AGCMs are typically forward-modelled, theoretically derived and validated against modern climate data before they can be applied (Annan and Hargreaves, 2015). However, given the complex nature of the water cycle, especially in the tropical-extratropical regions, where many factors influence vapour generation, transport and precipitation type, a backward propagation model that incorporates isotopic modifications in meteoric waters may work better. Capturing the water cycle at any stage will always have its challenges given the potential for vapour mixing, escalated isotopic fractionation resulting from minute temperature changes, series of post-depositional isotopic modification to meteoric waters and most importantly, speeding up of the water cycle in response to a global rise in temperatures (Kendall and McDonnell, 2012). Here, the isotopic composition of any sampled moisture already includes all these processes, which simply need to be resolved as back-propagated. Whereas deducing the same from the start (forward model) could fail to incorporate the dynamic nature of moisture transport.

GLIMPSE (GLobal Isotope Mapping and Precipitation Simulation Experiment) is an innovative model designed to simulate the global water cycle by integrating physics-based principles and machine learning algorithms. Employing satellite-derived and hydrological input parameters, GLIMPSE employs a 1-D moisture back-propagation schema to track the isotopic composition of surface waters back to their origin vapours, considering various precipitation sources such as rain, snow, glacier melt, groundwater, and evapotranspiration. The model introduces a novel moisture trajectory method, tracing Atmospheric Rivers from daily Total Precipitable Water Vapour (TPWV) gradients towards local precipitation sinks. Atmospheric Rivers can be defined as long, narrow, and transient corridors of strong horizontal water vapour transport, typically associated with a low-level jet stream ahead of the cold front of an extratropical cyclone (American Meteorological Society, 2017). The water vapour in atmospheric rivers is supplied by tropical and/or extratropical moisture sources. Atmospheric rivers frequently lead to heavy precipitation where they are forced upward, for example, by mountains or an ascent in the warm conveyor belt. Along this trajectory, the sink atmospheric water isotope signature is integrated between high to low TPWV sectors, back-propagating to its source vapour. The source vapour composition is then calculated independently and resolved into open water evaporation, canopy water evaporation, direct evaporation from bare soil, snow evaporation and transpiration. GLIMPSE not only incorporates the series of hydrological and atmospheric processes and their associated fractionations during rain-snow-vapour interconversion but also accounts for recycled moisture and mixed moisture signatures in vapour supply. The model is deployed for the Himalayan river basins, arguably one of the most complex climatological systems in the globe due to its intricate topography, multiple sources and types of precipitation, and anomalous cryospheric input, at 0.1 degree spatial and 1-day temporal resolution.

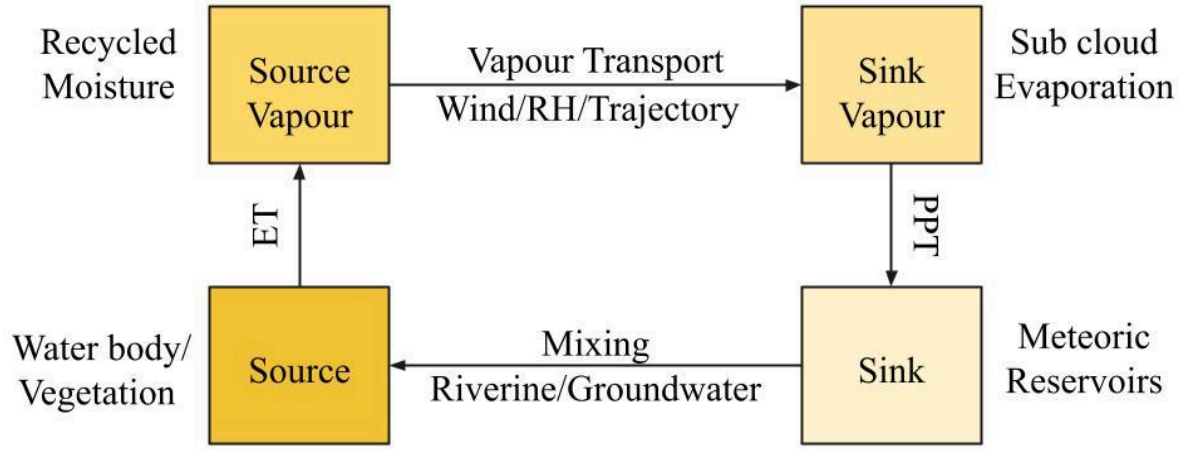


Figure 1: 1D pathway of the water cycle from source to sink.

2 Results and Discussion

GLIMPSE is a data-driven model built on observed or real-world water isotope data. Surface water isotopes ($\delta^{18}O$, $\delta^{2}H$ and d-excess) for Eastern Himalaya were compiled from 3 publications (Hoffmann et al., 2016; Voss et al., 2020; Dasgupta et al., 2021). The reasons behind choosing these 3 publications are (a) wide spatiotemporal coverage of a single river basin (i.e. Kosi), and (b) source and reservoir separation, i.e. data reported as snowpack, stream, tributary etc., sourced from rainfall, snowfall and groundwater. A total of 245 water samples, ranging from stream, river, snow, groundwater, glacier, and tributaries, spread over 3 seasons and an altitudinal gradient of 5 km has been compiled alongside climate and topographical data (Supplementary file df_spatial). All drivers were resampled to a common spatial (0.01*0.01 degree) and temporal (1 day) resolution.

The general outline of a backward propagation model (Figure 2) is taking the sampled water in a stream, river or lake and tracing it to the source vapour. This can be broken down into 3 steps. First, the meteoric reservoirs at the sink need to be isotopically characterised as a function of their local environmental drivers so that they can be modelled to achieve spatial congruence. This is done by means of a regressor model that predicts the $\delta^{18}O$ and $\delta^{2}H$ values of water on any given date and location. Second, categorise the source of modelled meteoric

waters as daily snowfall and rainfall for temporal congruence and compute the isotopic composition of the vapour from which said snow/rainfall originated. To do so, we identify the cloud height, cloud top temperature and rainout percentages, and apply a Rayleigh fractionation model unless precipitation occurs in subzero clouds, whereby we employ Majoube (1971). In step three, we identify the moisture pathway using atmospheric rivers for the sampled season and integrate the isotopic composition of water vapour from the sink to the source atmospheric conditions for infinitesimal changes in humidity, temperature, cloud height, wind pattern while accounting for progressive rain out and recycled moisture uptake (Merlivat and Jouzel 1979; Jouzel and Koster 1996; Cappa et al. 2003). In the following section, we will utilise in situ water isotope data from the Himalayan surface water inventories and satellite-derived climatological and topographic parameters to formulate sink meteoric water isotopic distribution, with the aid of exploratory data analysis, data clustering and feature importance, and gradient boosting. The sink isotope composition for meteoric waters will be traced to the sink-overhead atmosphere, where we simulate precipitation and condensation conditions. Next, the sink atmospheric vapour will be backpropagated to its source atmosphere alongside any changes brought on due to rainout and/or moisture recycling, along with a newly devised protocol for generating atmospheric rivers or moisture trajectory pathways. Finally, the vapour at the source will be resolved into its source components, such as open water evaporation, evapotranspiration and tertiary sources of water vapour. The source and transported isotopic composition of vapour will be computed at a seasonal scale following daily simulations, keeping Eastern Himalaya as the constant sink. Vapour sources and their isotopic composition will be validated for those seasons and trajectories for which published data (precipitation $\delta^{18}O$, $\delta^{2}H$) is available.

2.1 Exploratory data analysis (EDA)

Exploratory Data Analysis (EDA) scrutinises water isotope values from 245 locations across a mountain transect encompassing four seasons. It employs statistical summaries and

graphical representations to reveal trends and patterns, providing climatic variables crucial for explaining isotopic composition at any space-time point. The water isotope composition (e.g. $\delta^{18}O$) reflects its source, contribution to the meteoric reservoir, formation conditions, mass transfer, formation time, and transport degree. Each driver is individually analysed against $\delta^{18}O$, unveiling intrinsic relationships such as the direct proportionality between $\delta^{18}O$ and temperature represented by a 2nd-order polynomial (Supplementary Figure 1,2).

Next, we employ a single-blind test to cluster the $\delta^{18}O$ values of all 245 samples using the K-means supervised algorithm. K-means clustering is a form of unsupervised machine learning algorithm used to classify data into a user-defined number of groups, k. Once k is selected, the initial positions of the cluster centres (or centroids) would be initialised by randomly selecting k of the input data points as the centroid starting positions. Given that our starting database has water samples collected from 8 different meteoric reservoirs (Figure 3a), we run K-means with k = 8, and then check against the classes reported alongside the data in the literature for validation. Please note that the 8 classes are sourced from rain, snow or groundwater. Thus, we reclassify the 8 classes into 3: rain-fed, snow-fed and groundwater-fed (Figure 3b). The same exercise is carried out on the temporal scale by segregation of the isotopic signature of water by season. At the sink box in Figure 1, the observed water isotope data (df_spatial) include the following description:

a. Class - Sampled meteoric reservoirs (stream, tributary, river, snow, snowmelt, glacier melt, groundwater-sourced spring and groundwater-sourced tributary) reclassified into rain-fed, snow-fed and groundwater-fed.
b. Season - winter, spring, monsoon and post-monsoon.
c. Isotope - $\delta^{18}O$ and $\delta^{2}H$ value of the sample.
d. Spatial variables - latitude, longitude and elevation.
e. Temporal variables - temperature, rain (surface runoff), gw (subsurface runoff) and cryo (snowmelt).

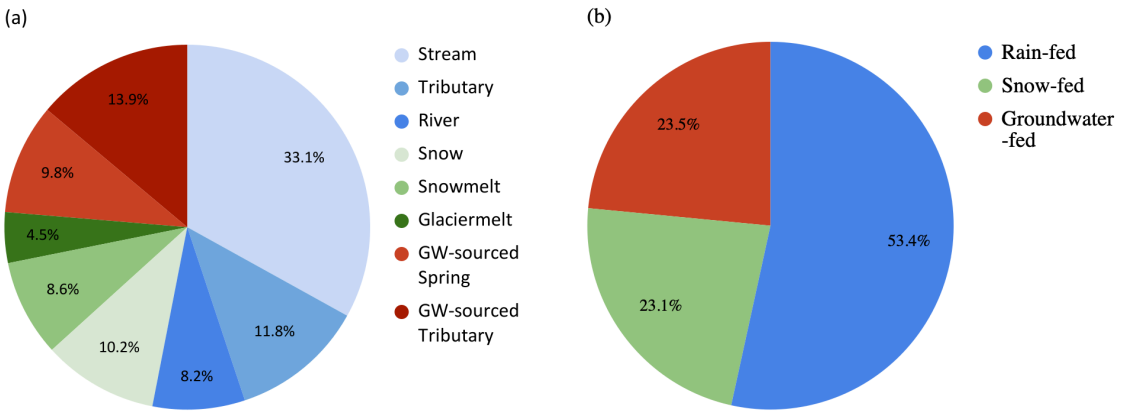


Figure 3: K-means clustering of (a) 8 moisture inventories for cross reference with reported classes of the samples into (b) 3 classes, i.e. rainfed, snow-fed and groundwater-fed meteoric water.

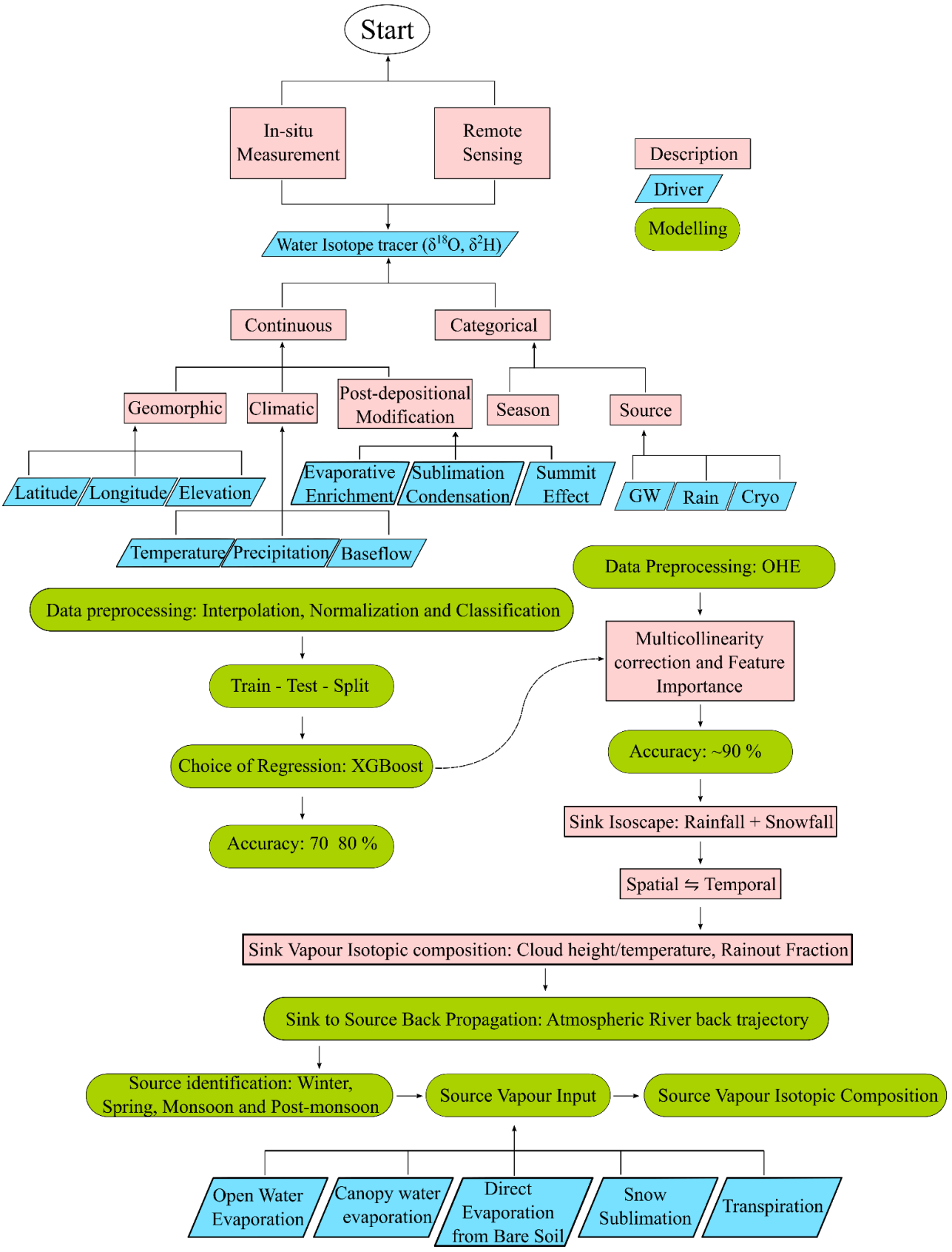


Figure 2: General outline of a backward propagation model from Sink to Source

2.2 Data Preprocessing

The model employs One Hot Encoding (OHE) for parameterising categorical features: class, source, and season—essential for model input. OHE transforms categorical values into binary vectors, representing each category with a binary array, wherein the index of the category is marked with 1 and others with 0s. This binary representation facilitates their integration into the model. Following data structuring and clustering, various regression models are explored to elucidate the relationship between target and feature variables. Linear and nonlinear polynomial regressors, decision trees, and gradient boost ML models are examined, with gradient boost algorithms exhibiting superior accuracy and minimal residuals (Table 1; Supplementary Figure 3). Specifically, the Extreme Gradient Boosting (XGBoost; Chen and Guestrin, 2016) algorithm proves noteworthy, preventing overfitting while optimising computational resources through additive training strategies and parallel computations. The XGBoost model is applied to a water isotope dataset, employing a 10-fold cross-validation for performance estimation. The XGB model's feature importance protocol, based on a set of Decision Trees, ranks the drivers (e.g., precipitation, evapotranspiration, LC). This ranking aids in model efficiency, workload reduction, avoidance of overfitting, and complexity reduction by incorporating only significant drivers, enhancing the δ18O prediction. Hyperparameters of the model are available with the code. The incorporation of feature importance contributes to refining model efficiency and selecting influential drivers, ensuring a more streamlined and impactful $\delta^{18}O$ prediction.

Table 1: A tabulated description of the different regressors applied to model $\delta^{18}O$ at Sink.

Methods:	Linear Regression	Polynomial Regression	Decision Tree	Gradient Boost
Accuracy (%)	68-70	50-70	80-90	92.6

The weightage of all the drivers or features on target variables ($\delta^{18}O$) were ranked following the XGB regressor feature importance protocol available on scikit-learn library. Feature importance refers to techniques that assign a score to input features based on their usefulness in predicting a target variable (Louppe et al., 2013; Strobl et al., 2007). Feature importance scores play a vital role in a predictive modelling project, including providing insight into the data, the model, and the basis for dimensionality reduction and feature selection that can improve the efficiency and effectiveness of a predictive model on the problem. Feature importance serves two purposes. First, to improve the model efficiency, reduce the workload, avoid overfitting and reduce the complexity of the model. Second, it allows us to incorporate only those drivers that have a significant impact on the output i.e. $\delta^{18}O$ prediction. XGB Feature Importance test ranks temperature, latitude and elevation as the top determining drivers for $\delta^{18}O$ prediction (Supplementary Figure 4).

2.3 Spatio-temporal modelling of water isotopes

After EDA and data preprocessing, GLIMPSE incorporates a modelling pipeline to predict the temporal $\delta^{18}O$ of source moisture based on key meteorological drivers. Here's a detailed summary of the code: The code begins by importing necessary libraries, including pandas, numpy, scikit-learn, and xgboost. It then loads the normalised and pre-processed data into a pandas DataFrame called `df_spatial`. The feature columns (Latitude, Longitude, Elevation, Temperature, and $\delta^{18}O$) are separated into `X`, while the target columns (Class and Season) are separated into `y`. The target columns are one-hot encoded to convert categorical variables into binary format. Additionally, these one-hot encoded columns are multiplied by the 'Temperature' column to remove multicollinearity (Elevation and Temperature are collinear). This operation introduces interaction terms between the temperature and target classes. These new columns are stored in `y_train_encoded` and `y_test_encoded`, respectively. After this, the 'Temperature' column is dropped from the feature set. Next, the

spatial data is split into training and testing sets using stratified sampling to maintain the proportion of classes in both sets. The training and testing sets are scaled using standardisation to centre the data around zero mean and unit variance. An XGBoost regressor model is then initialised and trained on the training data (`X_train_scaled` and `y_train_new`). The trained model is used to predict the $\delta^{18}O$ values on the test set (`X_test_scaled`), and the feature importances are calculated and printed, indicating the relative importance of each feature in making predictions. The code proceeds to read the temporal data from a new DataFrame called `df_temporal`. The temporal DataFrame is one-hot encoded to convert categorical variables into binary format. Like before, the one-hot encoded columns are multiplied by the 'Temperature' column to account for the interaction between temperature and targets. The XGBoost model is then used to make predictions on the temporal DataFrame for two sets of predictions: `y_pred_temporal_rain` and `y_pred_temporal_snow`. These predictions represent the temporal $\delta^{18}O$ values for rain and snow precipitation, respectively. To calculate the temporal $\delta^{18}O$ of precipitation, the code combines the predictions for rain and snow based on diurnal proportions. The `Sink Precipitation $\delta^{18}O$` is calculated as the weighted average of `y_pred_temporal_rain` and `y_pred_temporal_snow` based on the ratio of snowfall to the sum of snowfall and rainfall, and vice versa. The code then calculates the fraction of atmospheric vapour precipitated (F) as the ratio of Global Precipitation Measurement (GPM) to Precipitable Water Vapour (PWV) values from the temporal DataFrame. Fractionation factors (alpha) at the sink atmosphere and each source are calculated using temperature values from the temporal DataFrame. These fractionation factors represent the isotopic fractionation during the phase change of water vapour. Finally, the $\delta^{18}O$ of sink vapour and source vapours are calculated. $\delta^{18}O$ of sink vapour is obtained by subtracting the natural logarithm of (1 - F) divided by the fractionation factor (alpha) at the sink atmosphere. For each source, $\delta^{18}O$ of vapour is calculated based on the difference between the K term and the M term, where the K term represents the isotopic

fractionation due to the source, and the M term represents the isotopic fractionation due to kinetic processes (see section 2.5). The code also combines the $\delta^{18}O$_Vapour of all six sources as a function of their seasonality, using trajectory weights, to get the final result, `Source_Combined_Seasonally`. This value represents the combined temporal $\delta^{18}O$ of vapour from all six sources, weighted based on their respective contributions. In summary, the pipeline applies data preprocessing techniques, trains an XGBoost model on spatial data, and then uses the trained model to predict temporal $\delta^{18}O$ values of precipitation based on seasonal features and isotopic fractionation factors.

2.4 Source identification and moisture pathway

The atmosphere consists of freely flowing water vapour, which may or may not cause precipitation. The movement of the water vapour is along pressure gradients but is also influenced by wind movement, latitudinal and longitudinal circulation belts, humidity and temperature gradient. In the tropics, the Hadley and Walker circulation occur along the N-S and E-W directions, respectively, which resolve to a general pattern of water vapour movement in the NNW-SSE direction, where the source and sink of the moisture gradient depend on the time of the year and the relative positioning of landmass and oceans. We reimagined this vast body of water vapour and its potential to precipitate by calculating the flow direction and flow accumulation, following the same rationale of river drainage on land. The resultant atmospheric rivers map the Total Precipitable Water Vapour (TPWV) pathways for each season and each location on the planet.

The Flow Direction tool in ArcGIS Pro is used to derive the hydrologic characteristics of a watershed and determine the direction of flow from every cell in it (Rampi et al., 2014). This tool takes a surface as input and outputs a raster showing the direction of flow out of each cell. If the Output drop raster option is chosen, an output raster is created showing a ratio of the maximum change in TPWV from each cell along the direction of flow to the path length between centres of cells and is expressed in percentages. There are eight valid output

directions relating to the eight adjacent cells into which flow could travel. Each direction is the resultant of the pressure atmospheric gradient at contemporaneous altitude with the TPWV. In hydrology, this approach is commonly referred to as an eight-directional (D8; Figure 4) flow model and follows an approach presented by Jenson and Domingue (1988).

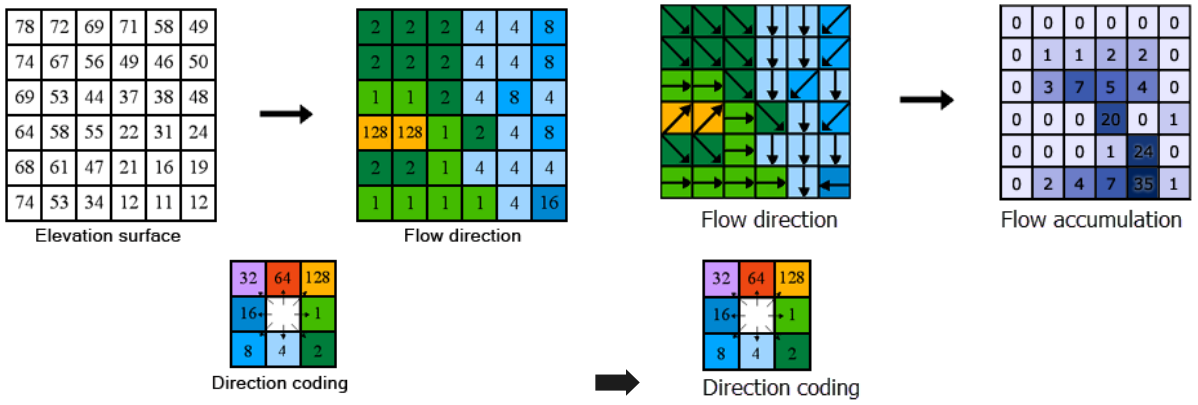


Figure 4: Flow direction and accumulation derivation on ArcHydro for TPWV. The left image shows the direction of travel from each cell, and the right image shows the number of cells that flow into each cell.

The Flow Accumulation tool calculates the accumulated flow of TPWV (Figure 5b) as the accumulated weight of all cells flowing into each downslope cell in the output raster. We apply 2 weight factors, Humidity and Wind direction (Figure 5a, 5c), to each cell, and the value of cells in the output raster is the number of cells that flow into each cell. Seasonal averages for TPWV, wind direction and humidity profiles were generated from satellite data, following which flow direction-accumulation tools were employed (Figure 5d). The resultant pathways, passing over the study area, were traced up to their source for each season. The model identified 6 pathways (Figure 5e). 1 each for monsoon and post-monsoon originating over central BoB (Source 1) and eastern BoB (Source 2), respectively; 2 each for spring and winter originating over central Nepal (Source 5) and southern Tibet (Source 6), and the northwestern Himalayas (Sources 3, 4) respectively.

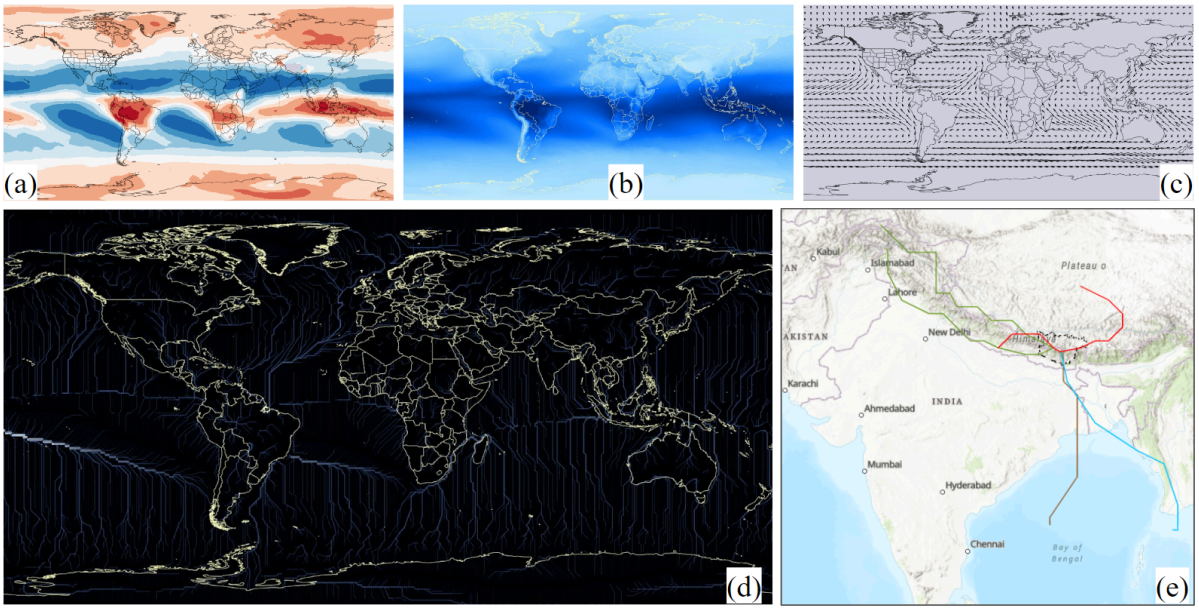


Figure 5: Generation of moisture trajectories using (a) Relative Humidity and (c) Wind direction limits on (b) TPWV flow-direction-accumulation to yield (d) atmospheric rivers. The seasonal trajectories for the eastern Himalayas are shown in (e) with winter (green), spring (red), monsoon (brown) and post-monsoon (blue) arms.

Once the moisture pathway is determined from the above exercise, we can calculate the amount and $\delta^{18}O$ value of source moisture w.r.t sink by integrating TPWV $\delta^{18}O$ value with limits at source and sink. The equation is empirically derived by budgeting moisture $\delta^{18}O$ value at sink vapour and at *dx* distance before sink vapour. Here, *dx* is represented at infinitesimal TPWV change between sink vapour and (sink vapour minus vapour at *dx* position before). The derivation is given below in 2.5. For the long-form integration please refer to the Supplementary file.

2.5 Backpropagation of vapour isotope composition along the delineated moisture pathway

Along the established trajectory from source vapour to sink vapour, consider an infinitesimal segment where the moisture parcel moves from C to C'

C (initial atmospheric conditions) → C′ (final atmospheric conditions)

Infinitesimal distance

$$AV_f = AV_i - RO\ (\%) + RM\ (\%)$$

where AV is Atmospheric Vapour (initial and final), RO and RM are Rainout and Recycled Moisture percentages.

Assuming isotopic compositions to be proportional to amount,

$$\delta^{18}O_{AVf} = \delta^{18}O_{AVi} - \delta^{18}O_{RO} + \delta^{18}O_{RM}$$

$$\delta^{18}O_{AVi} = \delta^{18}O_{AVf} + (\delta^{18}O_{RO} - \delta^{18}O_{RM})$$

Applying $R = R_0.f^{\alpha-1}$,

$\delta^{18}O_{AVf} = \delta^{18}O_{C'} \times f^{\alpha(C')-1} = \delta^{18}O_{C'} \times k_1.AV_{C'}.^{\alpha(C')-1}$

$\delta^{18}O_{RO} = (\delta^{18}O_{C'} \times \alpha_C) \times k_2.(AV)_{C'}^{\alpha(C')-1}$

$\delta^{18}O_{RM} = (\delta^{18}O_{AV} \times ET) + (\delta^{18}O_{RO} \times SCE) = [\delta^{18}O_{AV(k3)} \times ET] + [\delta^{18}O_{RO(k2)} \times SCE]$

where ET is evapotranspiration, SCE is sub-cloud evaporation and fractional atmospheric vapour, RO and RM relative to AV at C′ are k_1, k_2 and k_3 respectively.

$$\therefore\ \delta^{18}O_{AVi} = AV^{\alpha-1}.\ K.\ \delta^{18}O_{C'} - AV^{2(\alpha-1)}.M$$

where, AV is now replaced by *x* or the infinitesimal unit of integration,

$$K = (k_1 + k_2) \text{ and } M = k_3 + k_2.\delta^{18}O_{AV(C')}.\alpha_C$$

Summation of individual trajectories: $Traj_{total} = Traj_{CC'} + Traj_{C'C''} + Traj_{C''C'''} \ldots$

Integrating for cumulative AV

$$\delta^{18}O_{Source} = \int x_i^{\alpha(i)-1}.K.\delta^{18}O_{Sink}.dx - \int x_i^{2[\alpha(i)-1]}.M.dx$$

$$\therefore\ \delta^{18}O_{Source} = K.[\delta^{18}O_{Sink}.\ (AV_{Sink}^{\alpha(sink)} - AV_{Source}^{\alpha(source)})/(\alpha_{sink} - \alpha_{source})]\ -$$

$$M.[(AV_{Sink}^{2\{\alpha(sink)-1\}} - AV_{Source}^{2\{\alpha(source)-1\}})/2(\alpha_{sink} - \alpha_{source})] \qquad \text{(equation 1)}$$

2.6 Model Validation and Uncertainty

For validation purposes, the modelled isotopic composition of the source vapour (Equation 1) was compared with actual measurements (Figure 6a). For the Himalayan run, this was made possible by comparing the IAEA station of Ghalegaon, where continuous rainwater isotope measurements were carried out by the Université de Lorraine-CNRS from 2016 to 2017. The $\delta^{18}O$ values modelled by GLIMPSE for water vapour generating over central Nepal (Source 6) and feeding precipitation events in the Kosi basin were compared with Ghalegaon daily $\delta^{18}O_{GNIP}$ data. The model was run with a 48-hour delay relative to the date of precipitation in Kosi to accommodate for the back trajectory period. This period may change depending on the moisture travel pathway, and therefore, the model is iterable between 0 hours upto 240 hours. The reasoning behind choosing 48 hours was the accuracy of fit and the short distance covered by the spring trajectory. The comparison shows a significant fit with T-Test score = 25 at 95 % and Pearson correlation = 0.7 at $p < 1x10^{-5}$, therefore validating the spring arm of the model. For the remainder seasons, similar station-observed data is required.

One of the key aspects of GLIMPSE is the conversion of spatial information to a temporal scale. As the input dataset is discrete and exhaustive, and represents specific sampling days of the labelled season, they do not constitute a time-continuous series over 365 days. As an alternative, the spatial relationship that the water isotopes have with its drivers is exploited. Drivers such as temperature, and the relative contribution of rain and snow over space can be resolved in time for spatiotemporal congruence. In principle, this works well but succumbs to the artefacts of the spatial data distribution. As evident in Figure 6b, the sink rain and snow $\delta^{18}O$ values modelled after discrete spatial values do not vary naturally as their vapour counterparts, which have been modelled with atmospheric variables that vary with

time. Apart from random error associated with input data and satellite variables, this conversion is a significant caveat of the model and completely depends on data assimilation.

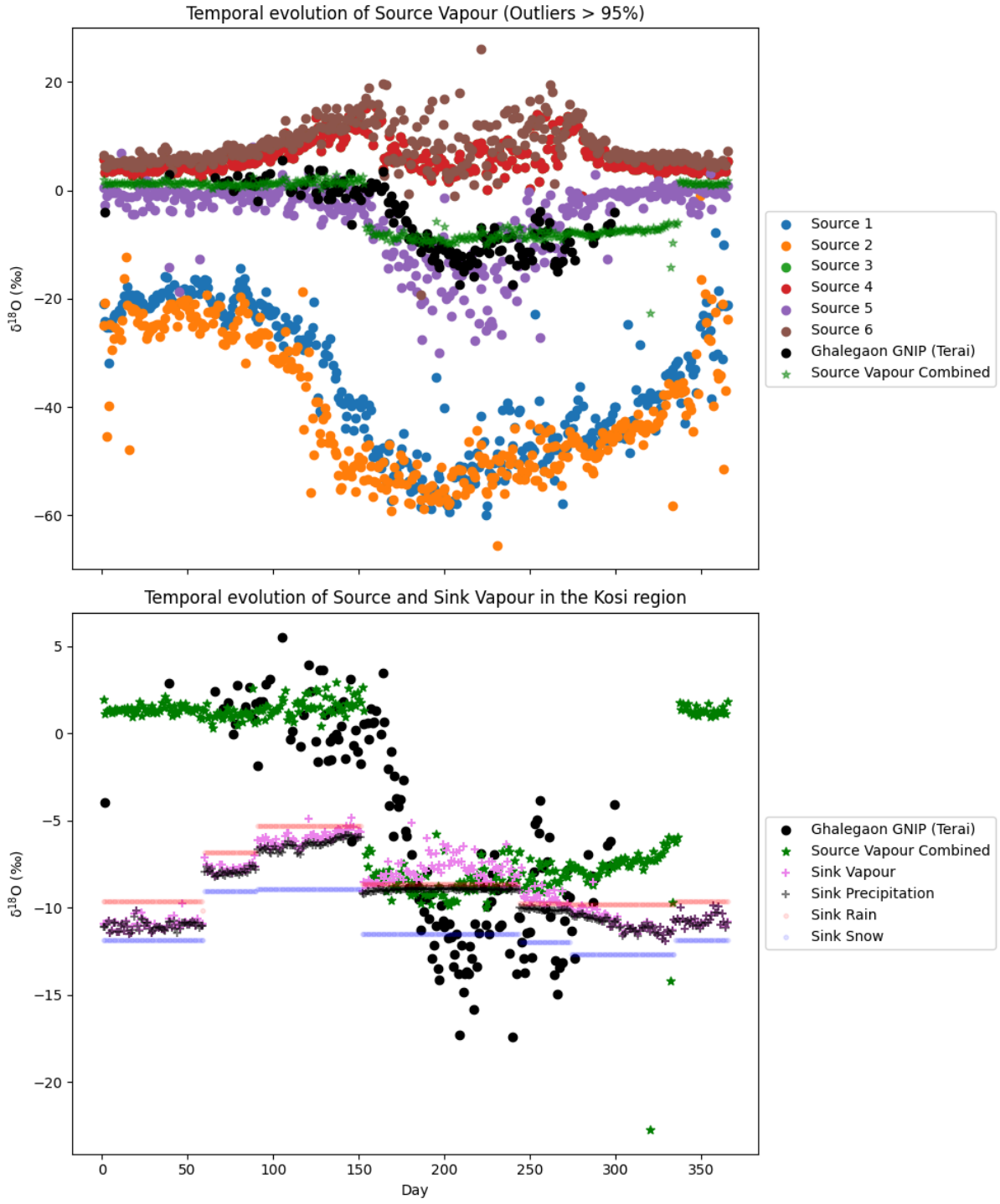


Figure 6: Validation of source moisture $\delta^{18}O$ values with IAEA GNIP $\delta^{18}O$ values for spring trajectory in Ghalegaon, central Nepal.

While it is true that models based on satellite-derived geophysical variables incorporate not only the inherent error in individual datasets, but also their combined error resultant of mathematical propagation, we have tried to minimise such random errors wherever possible. An ensemble method for error propagation has been employed, where the reported error in each measurement, sampled and simulated, are algebraically added onto the subsequent node of the 1D pathway. The error associated with each remotely sensed driver is available in the supplementary section.

2.7 Implications: Source, spatial and temporal variation in water isotopes

The **isotopic variability of rainfall** is modulated by the interaction between ambient vapour and raindrops, influenced by source moisture (Sinha, 2020). Coastal areas exhibit stable isotopic compositions controlled by continuous extraction from the atmospheric vapour reservoir, while inland areas and mountain regions show influence from the recycling of continental moisture (Njitchoua, 1999). Peng (2011) found in Taiwan that topographic type, temperature, and rainfall affect transpiration, a key component of moisture recycling, using a water isotope approach. The Intertropical Convergence Zone's (ITCZ) movement significantly influences moisture source variability in the Himalayas, with the Indian summer monsoon intensity linked to the ITCZ dominating precipitation $\delta^{18}O$ (Cai, 2012). Sanwal (2013) showed that the ITCZ's position affects precipitation in the Central Indian Himalayas. Accurate determination of **moisture sources** is pivotal for comprehending precipitation patterns in montane regions. Bottyán (2017) and Winschall (2013) stress the significance of identifying these sources, particularly emphasising stable isotope analysis. Gimeno (2020) and Ordoñez (2012) underscore the importance of this knowledge in predicting rainfall distribution and understanding moisture transport in extreme precipitation events. The challenges in determining moisture sources and precipitation in a montane basin are multifaceted, with the basin's small size and significant variability posing difficulties in obtaining reliable data (Kostka, 1994). The impact of reduced data networks on rainfall

estimates further complicates this issue, as discussed by Arora (2006). GLIMPSE stands as a robust solution for these challenges in montane hydrology. The model, utilising surface hydrology, stable isotope networks, and meteorological conditions, proficiently breaks down moisture source pathways identified from atmospheric rivers on individual days (Figure 5e). In the eastern Himalayas, the primary moisture source is the monsoonal air masses originating from the Bay of Bengal and transported along the Bengal-Brahmaputra basin (Hren, 2009). Around 80% (1500 mm) of the precipitation in this region occurs during the monsoon period of June to September (Dasgupta et al. 2021). Non-monsoonal precipitation accounts for the remaining 20%, which comprises recycled moisture and Winter westerly disturbances (WWD) sourced precipitation. WWD are characteristically eastward propagating extratropical cyclones with a significant reduction in precipitation amount from west to east (Lang and Barros 2004; Barlow et al. 2005), resulting in most of the precipitation in the east during non-monsoon months being locally sourced, i.e. recycled. Recycled moisture influences Tibetan precipitation, ranging from 37% in the Qilian Mountains during the plant growing season (Qiu, 2021) to nearly half in the northwestern Tibetan Plateau (An, 2017). This is particularly crucial considering the role of Tibetan Plateau ecosystems in contributing to local and remote precipitation through moisture recycling (Li, 2022). Aizen (1996) identified the Indian and Pacific Oceans, as well as the Atlantic Ocean, as sources of moisture on the northern slope of the Himalayas and southeastern Tibet, which does not seem to be the case in our study. Bohlinger (2017) adds that extreme precipitation events in Nepal are associated with mid-level troughs and low-pressure systems, with a significant contribution from increased (recycled) moisture uptake over the Indo-Gangetic Plain during early spring and pre-monsoon months. In summary, each of the 6 moisture sources determined by the seasonal trajectories can be cross-referenced with meteorological studies covering ISM, WWD and locally sourced moisture.

Besides the source and spatial variability of moisture, the **temporal** evolution of precipitation isotopes needs attention. Each seasonal arm of the model brings in moisture from a different source (open water evaporation, canopy water evaporation, direct evaporation from bare soil, snow evaporation and transpiration), directed by different atmospheric conditions and resulting in a different range of isotopic values (Supplementary Figure 5). These differences are more prominent during the warmer and wetter months, as seen in Figure 6a (Sources 1 and 2), but taper to closer values in winter/spring (Sources 3-6), suggestive of a contrasting isotopic signature of moisture sourced from the open ocean and land respectively. The model is validated by its spring trajectory sourced from central Nepal (Source 5), where existing IAEA precipitation $\delta^{18}O$ records concur with modelled source vapour $\delta^{18}O$ values. The seasonally combined vapour $\delta^{18}O$ values (Figure 6b) are approximately 1-2 ‰ lower relative to Ghalegaon daily $\delta^{18}O_{GNIP}$, with both datasets showing lower (< -5 ‰) isotopic values in monsoon months than non-monsoon months (-5 to -20 ‰). This is consistent with areas that receive the majority of their rain from monsoon systems, and their stable isotope amount effect is often invoked to explain patterns of isotopic composition of rainfall in the tropics. Higher values of $\delta^{2}H$ and $\delta^{18}O$ were observed during months with less rainfall, and lower values of $\delta^{2}H$ and $\delta^{18}O$ correlated with higher precipitation amounts (Dansgaard, 1964). In terms of the rainfall amount effect, the content of isotopes in precipitation has a negative correlation with the rainfall because as the precipitation process continues, the content of heavy isotopes will decrease. An interesting feature of spatial imprint on the temporal variability of precipitation isotopes can be gleaned from the unique partitioning of moisture sources over the Himalayas based on evapotranspired amount and inventory for the source grid cell and the modelled source isotopic composition of vapour (Figure 7). Of the 3 clusters that formed, the first one appears over winter to early spring with ET < 1 kg m^{-2} day^{-1} and $\delta^{18}O$ composition ranging 0 to 2.5 ‰. This cluster represents a significantly lower moisture input along the winter and spring

pathways, sourced terrestrially (Sources 3-6) from the sum total of canopy evaporation, snow evaporation and transpiration, and isotopically enriched. The second and third clusters form over pre to post-monsoon months with ET ranging from 2 to 3.5 and 3 to 4 kg m^{-2} day^{-1} and $\delta^{18}O$ composition ranging 0 to 4 ‰ and -20 to -40 ‰ respectively, sourced primarily over open waters (Sources 1,2) and picking up moisture along path from the direct evaporation from bare soil. This essentially demonstrates the spatial identity or source imprint of precipitation on its temporal variability.

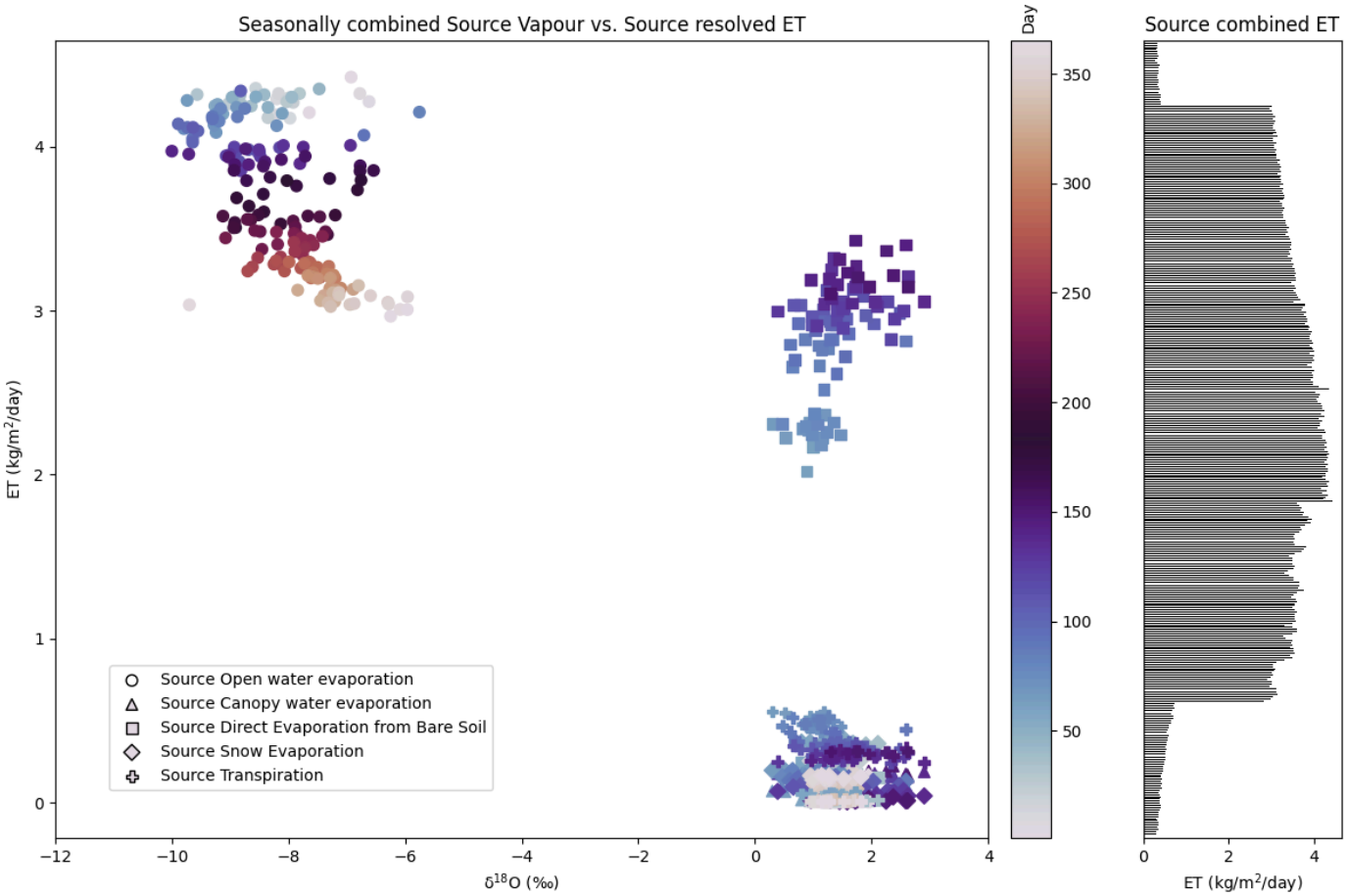


Figure 7: Linking source evapotranspiration inventories: resolved (left) and combined (right) with source vapour $\delta^{18}O$ values combined seasonally.

3 Concluding Remarks

The available range of satellite-observable geophysical variables that relate to the process of precipitation – such as evapotranspiration, air temperature, and net radiation – is

continuously growing and the quality of these data sets is constantly improving. As a result, models aiming at the accurate estimation of atmospheric circulation from satellite observations need to be updated to optimally incorporate these new data. Concurrently, as our knowledge of the relevant physical processes advances based on new evidence, these data-driven models should aim to increase their realism. With the overarching goal of improving our understanding of the global hydrological cycle, GLIMPSE is presented in this paper. Three major strides are achieved: (1) resolving the isotopic signatures of different meteoric reservoirs into the snow and rain-fed proportions using ML, (2) constructing a sink to source daily moisture pathways using a novel TPWV flow accumulation method, and (3) back-propagation of water vapour along these pathways to their respective terrestrial and open water sources. Using GLIMPSE, one can utilise the surface water isotopic composition sampled or reported by studies and trace it to its origin vapour. In doing so, it incorporates the series of hydrological and atmospheric processes and their associated kinetic fractionations and accounts for signatures of recycled and mixed moisture. The strength of the model lies in its subparts. Unlike existing studies, GLIMPSE uses the potential of ML to cluster meteoric water isotopic composition into its precipitation source and therefore develop functional relationships in the form of isoscapes for the sink region i.e. Kosi Basin, for a given time frame (e.g. month or season). Both on-field sampled inventories (e.g. rain, snow, groundwater etc.) and those derived from the single-blind K-means suggest a strong classification scheme stored in the $\delta^{18}O$ - $\delta^{2}H$ signature of surface waters. The second point is the level of accuracy achieved by XGB over traditional methods to simulate a meteoric water isoscape in the Kosi basin. Next, GLIMPSE works by tracking moisture through atmospheric river-directed pathways instead of the ordinarily used wind back trajectories (e.g. HAD30, HYSPLIT). Finally, GLIMPSE's ability to integrate the $\delta^{18}O$ value total atmospheric water vapour along the trajectory path for infinitesimal changes in atmospheric conditions, eventually being calculated as its isotopic evolution.

In this study, we model surface water isotope composition, cluster them on the basis of snow or rain-fed precipitation, project the cloud conditions which generated such precipitation and then backpropagate this cloud parcel to its source vapour for each season's atmospheric river-dominated moisture path. The results for the Himalayan run were verified against rainfall isotope measurements at IAEA GNIP stations. GLIMPSE ran for the eastern Himalayas, configured the source, transport length and path and mode of precipitation for the eastern Himalayan surface waters. It also helped resolve the relative contribution of oceanic and terrestrial vapour sources, both as open-water evaporation and open-water evapotranspiration. Following the advances in satellite technology and the increased availability of new data, GLIMPSE will be further optimised in the coming years. Current activities concentrate on the incorporation of new constraints on sub-cloud evaporation, the application of higher resolutions and near-real-time TPWV, and the improved partitioning of vapour to snow and vapour to rain fractionation at the sink. Given that this model is derived on publicly sourced and remotely sensed data, it is open for access, modifications and re-runs in different geographies with differing precipitation systems. Future optimizations include incorporating new constraints, higher resolutions, and near-real-time TPWV, making GLIMPSE adaptable and accessible for different geographies and precipitation systems.

Reference